Q3	Configure file and directory permissions (chmod / chown)	5 Marks

Objective: Restrict access to sensitive files using ownership and permission bits, and verify the change.

Step-by-Step Algorithm

Step	Phase	Action	Command / Tool	What You Should See
1	SETUP	Identify a sensitive file that needs restricted access.	ls -l sensitive.conf	Current permissions
2	CONFIGURE	Set the owner and group appropriately.	`sudo chown root:secteam sensitive.conf`	Ownership changed
3	CONFIGURE	Apply restrictive permission bits.	`sudo chmod 640 sensitive.conf`	Mode = -rw-r-----
4	TEST	Re-check the permissions to confirm.	ls -l sensitive.conf	New permissions shown
5	TEST	Test that an unauthorised user is denied.	`sudo -u nobody cat sensitive.conf`	Permission denied

Phase key: ● SETUP · ● CONFIGURE · ● EXECUTE · ● TEST · ● ANALYSE · ● REPORT · ● FINISH

Worked Example (with sample values & output)

Applying the steps to lab-vm

The configuration file starts world-readable. After changing ownership and applying mode 640, only root and the secteam group can read it, and other users are denied.

Stage	Owner:Group	Permissions	Who can read
Before	user:user	-rw-r--r--	Everyone
After	root:secteam	-rw-r-----	root + secteam only

Sample output:

```
$ ls -l sensitive.conf
-rw-r----- 1 root secteam 220 Mar 5 sensitive.conf
```

Reusable Template

Permission-change record template

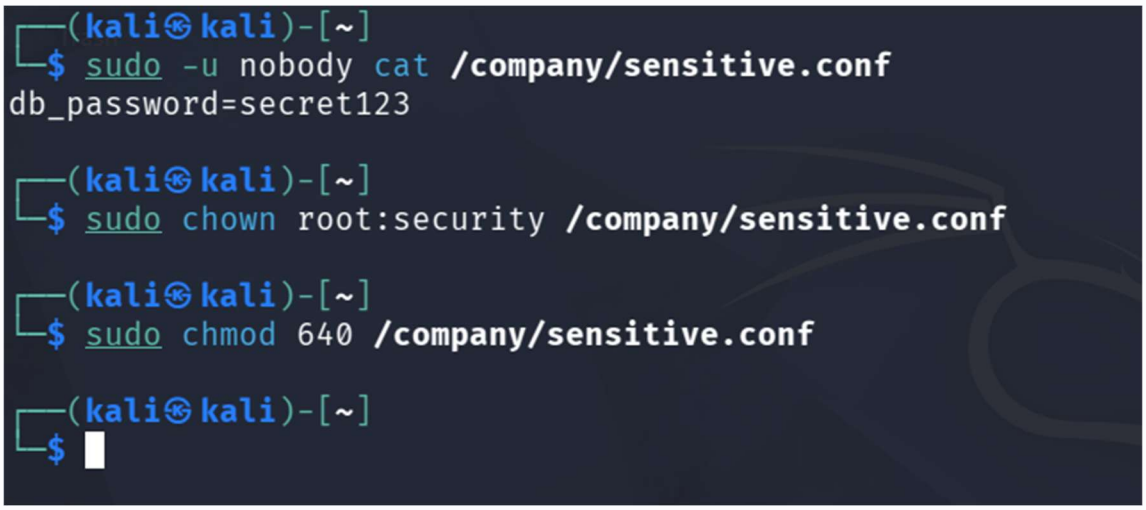
File	Before (owner / mode)	After (owner / mode)
/company/sensitive.conf	root:root, -rw-r--r--	root:security, -rw-r-----
/company/hr/salary.txt	root:root, -rw-r--r--	root:hr, -rw-r-----

Your Output / Screenshot (paste result here):

1. I created a file sensitive.conf



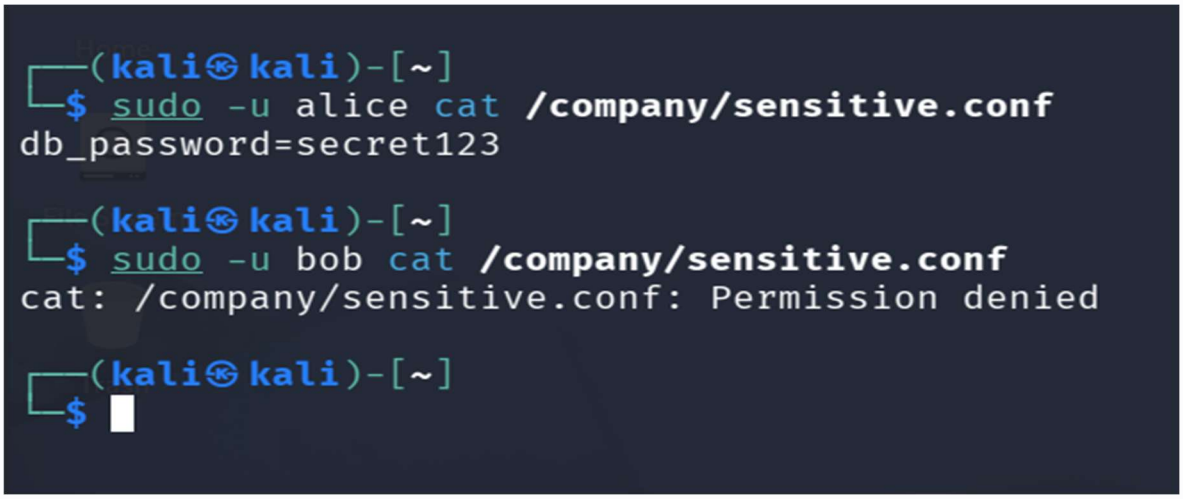
2. Step02 and 03 set owner and group and permissions



3. Recheck permissions and unauthorised user denied



4. Authorised vs unauthorised group members



5. Applying ownership and permissions

```
┌──(kali㉿kali)-[~]
└─$ echo "Employe salary data" | sudo tee /company/hr/salary.txt
Employe salary data
```

```
┌──(kali㉿kali)-[~]
└─$ ls -l /company/hr/salary.txt
-rw-r--r-- 1 root root 20 Oct 10 00:55 /company/hr/salary.txt
```

```
┌──(kali㉿kali)-[~]
└─$ sudo chown root:hr /company/hr/salary.txt

┌──(kali㉿kali)-[~]
└─$ sudo chmod 640 /company/hr/salary.txt

┌──(kali㉿kali)-[~]
└─$ ls -l /company/hr/salary.txt
-rw-r----- 1 root hr 20 Oct 10 00:55 /company/hr/salary.txt
```

6. Test denied and allowed

```
┌──(kali㉿kali)-[~]
└─$ sudo -u nobody cat /company/hr/salary.txt
cat: /company/hr/salary.txt: Permission denied

┌──(kali㉿kali)-[~]
└─$ sudo -u david cat /company/hr/salary.txt
Employe salary data
```